Algebra SEP Worksheet 2 7/17/25

Exact Sequences

Pre-Heat

Exercise 1. Let G, H, and K be groups. Prove that

(a) The sequence $1 \to H \xrightarrow{f} G$ is exact if and only if f is injective.

(b) The sequence $G \xrightarrow{h} K \to 1$ is exact if and only if h is surjective.

(c) The sequence $1 \to H \xrightarrow{f} G \xrightarrow{h} K \to 1$ is exact if and only if f is injective, h is surjective, and h induces an isomorphism $K \simeq G/\operatorname{im}(f)$.

Exercise 2. Let $0 \to A \xrightarrow{f} B \xrightarrow{g} C \to 0$ be an exact sequence of abelian groups.

(a) Prove that there are functions (not necessarily homomorphisms) $r : B \to A$ and $s : C \to B$ such that $r \circ f$ and $g \circ s$ are the identity functions on A and C respectively. The function r is called a retraction, and the function s is called a section.

(b) Let s be a section. Prove that the function $A \times C \to B$ given by $(a, c) \mapsto f(a) + s(c)$ is a bijection.

(c) Give an example of a short exact sequence of abelian groups where B is not isomorphic to $A \times C$.

Exercise 3.

(a) Let $0 \to U \to V \to W \to 0$ be an exact sequence of finite dimensional vector spaces. Prove that $\dim V = \dim U + \dim W$.

(b) Let $1 \to H \to G \to K \to 1$ be an exact sequence of finite groups. Show that $|G| = |H||K|$.

The two above statements are equivalent to standard facts in linear algebra and group theory respectively (do you know which facts?) Can you give proofs that only rely on facts about exact sequences?

Exercise 4. Let $1 \to A \to G \to K \to 1$ be an exact sequence of groups, where A is an abelian group.

(a) Because A is (identified with) a subgroup of G, the elements of G act on A by conjugation (i.e., each element of $g \in G$ gives a map $A \to A$ by $a \mapsto gag^{-1}$.) Show that if two elements of G map to the same element of K, then they give the same action on A. Conclude that there is a well defined homomorphism $K \to \operatorname{Aut}(A)$.

(b) Let p and q be prime numbers, and suppose $|A| = p$ and $|K| = q$. Show that if $q \nmid (p-1)$, then $G \cong \mathbb{Z}/pq$.

Exercise 5. This question gives an example where exact sequences may have been relevant to your previous study of group theory.

(a) Let G be a group. The action of G on itself by conjugation gives a homomorphism $G \to \operatorname{Aut}(G)$. Show that the kernel of this homomorphism is the center $Z(G)$ and there is therefore an exact sequence

$$1 \to Z(G) \to G \to \operatorname{Aut}(G).$$

Note: the image of G in $\operatorname{Aut}(G)$ is called the inner automorphism group and denoted $\operatorname{Inn}(G)$, so that there is a short exact sequence

$$1 \to Z(G) \to G \to \operatorname{Inn}(G) \to 1.$$

(b) Suppose that $\operatorname{Inn}(G)$ is cyclic. Show that G is abelian.

Remark: If we define $\operatorname{Out}(G) = \operatorname{Aut}(G)/\operatorname{Inn}(G)$, the exact sequence in part (a) can be extended to the long exact sequence

$$1 \to Z(G) \to G \to \operatorname{Aut}(G) \to \operatorname{Out}(G) \to 1.$$

Cooking

Exercise 6. (Inspired by January 2019 Problem 1b) Recall that if A and B are abelian groups, then $\operatorname{Hom}(A, B)$, the set of homomorphisms from A to B, is also an abelian group. Furthermore, if $f : A_1 \to A_2$ is a homomorphism, then the map $f_* : \operatorname{Hom}(B, A_1) \to \operatorname{Hom}(B, A_2)$ defined by $g \mapsto f \circ g$ is also a homomorphism. Let

$$0 \to A_1 \to A_2 \to A_3 \to 0$$

be an exact sequence of A-modules.

(a) Prove that

$$0 \to \operatorname{Hom}(B, A_1) \to \operatorname{Hom}(B, A_2) \to \operatorname{Hom}(B, A_3)$$

is exact.

(b) Give an example where the exact sequence in part (a) cannot be extended to a short exact sequence.

Exercise 7. Let
$$0 \to A \xrightarrow{f} B \xrightarrow{g} C \to 0$$
be a short exact sequence of abelian groups. Prove that the following are equivalent:

(a) There exists a homomorphism $r : B \to A$ such that $r \circ f = \mathrm{id}_A$.

(b) There exists a homomorphism $s : C \to B$ such that $g \circ s = \mathrm{id}_C$.

(c) There is an isomorphism $h : B \to A \oplus C$ such that the diagram

$$\begin{tikzcd} 0 \arrow[r] & A \arrow[r, "f"] \arrow[dr, "\iota_A"'] & B \arrow[r, "g"] \arrow[d, "h"] & C \arrow[r] & 0 \\ & & A \oplus C \arrow[ur, "\pi_C"'] & & \end{tikzcd}$$

commutes. (That is, $\iota_A = h \circ f$ and $\pi_C = g \circ h^{-1}$.)

Exercise 8. Show that the previous exercise fails if we no longer require the groups to be abelian. Specifically, find an example where (b) holds but (a) and (c) do not.

Exercise 9. Let E be an abelian group, and suppose E has a subgroup A isomorphic to $\mathbb{Z}/9\mathbb{Z}$. Suppose further that the quotient $C = E/A$ is isomorphic to $\mathbb{Z}/9\mathbb{Z}$. That is, we have an exact sequence
$$0 \to \mathbb{Z}/9\mathbb{Z} \to E \to \mathbb{Z}/9\mathbb{Z} \to 0.$$

(a) Let $q\colon E \to C$ be the quotient map. Let $e_0, \dots, e_8 \in E$ be elements such that $q(e_n) = n$. (That is, choose a set of coset representatives.) Show that for each n, $9e_n \in A$.

(b) Show that if e'_n is a different coset representative for n, i.e. a different element so that $q(e'_n) = n$, then $9e_n = 9e'_n$.

(c) The above shows that the function $\mathbb{Z}/9\mathbb{Z} \to \mathbb{Z}/9\mathbb{Z}$ given by $n \mapsto 9e_n$ is well-defined. Show that this function is also a group homomorphism.

(d) The previous parts can be rolled into one big function

$$\begin{array}{ccc} \left\{ \begin{array}{c} \text{short exact sequences} \\ 0 \to \mathbb{Z}/9\mathbb{Z} \to E \xrightarrow{q} \mathbb{Z}/9\mathbb{Z} \to 0 \end{array} \right\} & \to & \mathrm{Hom}(C, A) \\ (0 \to \mathbb{Z}/9\mathbb{Z} \to E \xrightarrow{q} \mathbb{Z}/9\mathbb{Z} \to 0) & \mapsto & (n \mapsto 9q^{-1}(n)). \end{array}$$

Show that this function is a bijection by constructing an inverse. That is, given a homomorphism $\phi\colon C \to A$, construct an abelian group E fitting into such an exact sequence such that $9q^{-1}(n) = \phi(n)$. (Hint: let the underlying set of E be $A \times C$, and write down what the addition should be to satisfy this property.)

Exercise 10. In the previous question, we determined the number of short exact sequences $0 \to \mathbb{Z}/9\mathbb{Z} \to E \to \mathbb{Z}/9\mathbb{Z} \to 0$, which is already not an obvious thing. One thing to be pointed out is that this *does not* immediately give the number of different abelian groups E that can go in the middle. The reason is that some groups E get counted more than once when we count the exact sequences, because different maps $\mathbb{Z}/9\mathbb{Z} \to E$ into the same E get counted separately.

(a) Concretely, suppose $E = \mathbb{Z}/81\mathbb{Z}$, and consider the two maps $i_1, i_2 \colon \mathbb{Z}/9\mathbb{Z} \to \mathbb{Z}/81\mathbb{Z}$ given by $i_1(m) = 9m$ and $i_2(m) = 18m$. Show that the maps $\phi_1, \phi_2 \colon C \to A$ are not the same. Thus, this one isomorphism type has been counted at least twice (possibly you can already see how many times it has been overcounted).

(b) On the other hand, we can use our technology from above to count the number of non-isomorphic Es. Show that if $i_1, i_2 \colon \mathbb{Z}/9\mathbb{Z} \to E$ are two different maps so that the quotient $E/\operatorname{im}(i_1) \cong E/\operatorname{im}(i_2) \cong C$, and $\phi_1, \phi_2 \colon C \to A$ are the corresponding homomorphisms, then $\operatorname{im}\phi_1 = \operatorname{im}\phi_2$.

(c) How many non-isomorphic E are there? What are they?

Exercise 11.

(a) What, if anything, would change in our work from the previous two problems if instead $A = \mathbb{Z}/3\mathbb{Z}$? How many such E are there now?

(b) What, if anything, would change in our work if instead $A = \mathbb{Z}/27\mathbb{Z}$? How many such E are there now?